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DECEMBER 2025 |
Data Sci Final Presentation

THE CAPITAL INVESTOR.

Optimizing Airbnb Portfolio Strategy in Washington D.C.

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THE PROBLEM

- Saturated D.C. short-term rental market.
- Overpaying for investments.
- Need for ROI prediction.

THE BLIND INVESTING PARADOX

Decisions rely on broker anecdotes, past 'similar' deals, and lagging market comps — not predictive data.

We overpay for 'trophy' properties near the Mall and systematically miss undervalued assets in emerging neighborhoods. We need a way to know expected revenue before we sign the contract.

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OUR SOLUTION: “THE CAPITAL INVESTOR” PLATFORM

moving from **intuition & advice** to **standardized, repeatable, data-driven thinking**.

Strategic Dashboard (Power BI)

Answers: *What is happening in the market?*

AI Valuation Engine (Streamlit + Random Forest)

Answers: *What will happen for this specific property?*

STREAMLIT APP

GOOGLE COLAB

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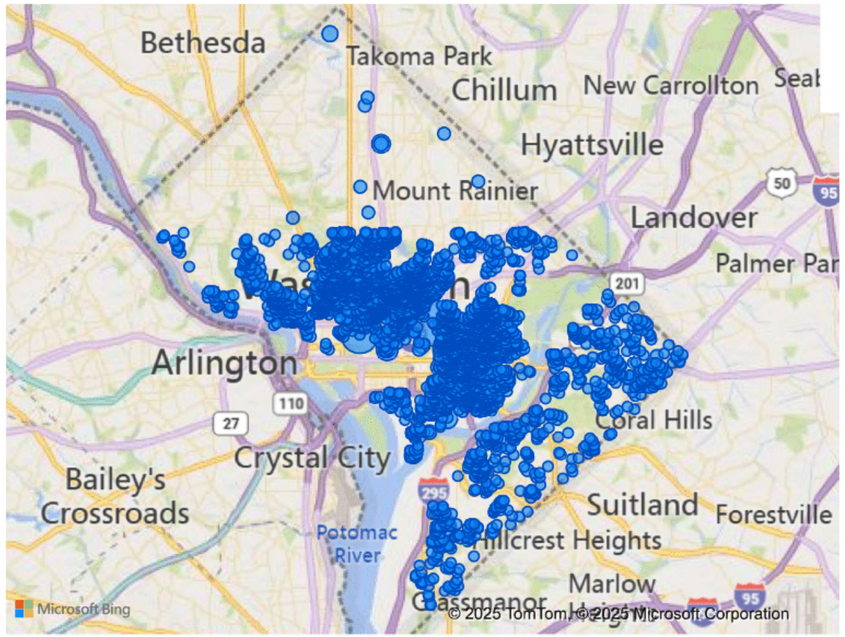
2.58
Average of dist_to_mall

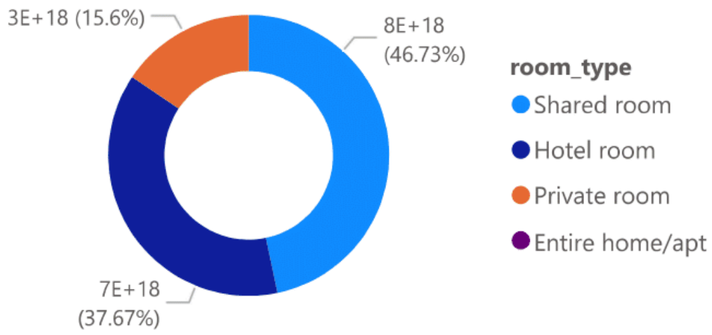
490
Count of price

This visual type is being retired soon. Upgrade now to avo... Upgrade map

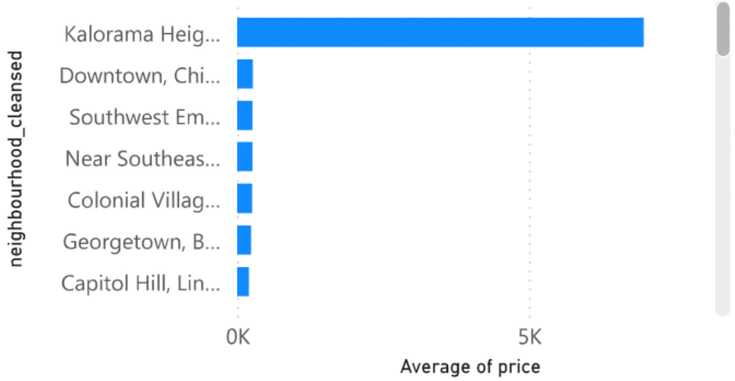
Count of price by latitude and longitude

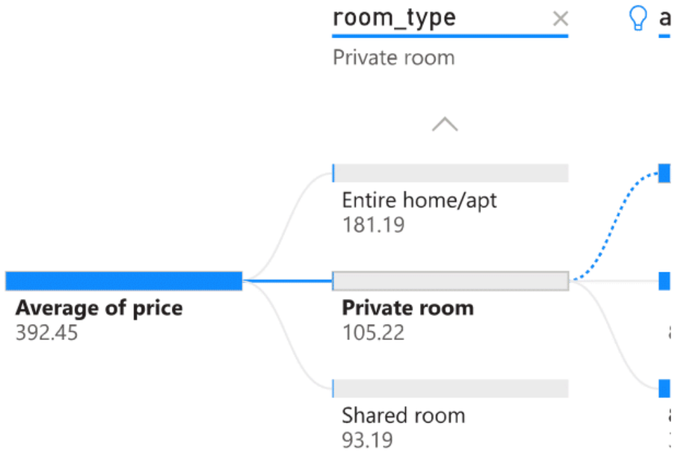


Sum of Total Active Listings by room_type

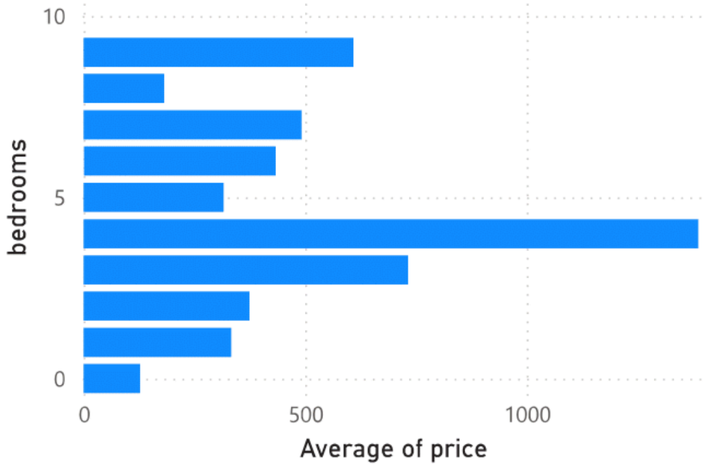


Average of price by neighbourhood_cleansed





Average of price by bedrooms



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THE TURNING POINT INSIGHT: THE “GOLDILOCKS ZONE”

< 1 MILE	Prices are high, but so are acquisition costs. The yield compresses.
>2.5MILES	Acquisition is cheap, but occupancy and pricing fall off sharply.
1.0–2.5MI	This is the **Goldilocks Zone** – not too expensive to buy, not too far to lose demand.

(DISTANCE FROM THE NATIONAL MALL)

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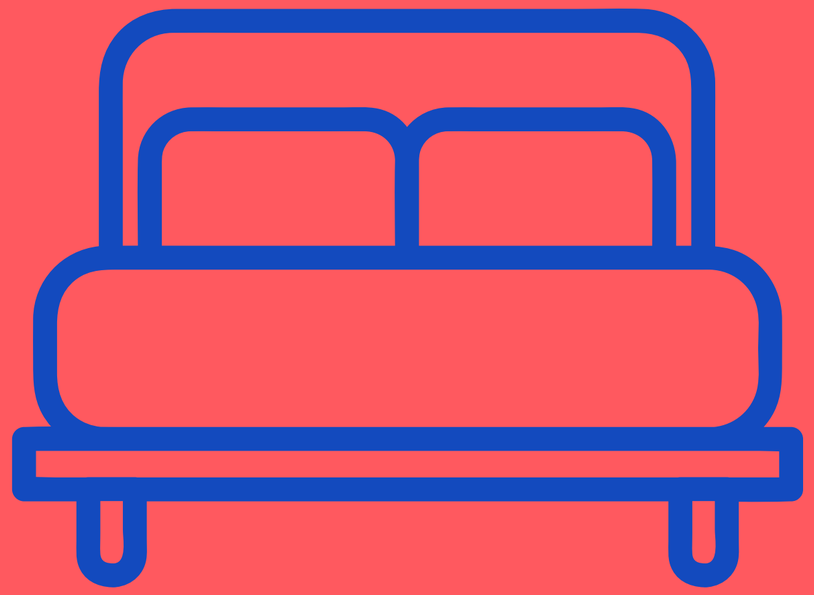
THE BEDROOM SWEET SPOT



LOOKING AT PRICE VS. BEDROOM COUNT

In the Goldilocks Zone, we should prioritize 2-bedroom condos over larger units.

JUMPING FROM 1 → 2 BEDROOMS

Increases nightly revenue by about $**150**

JUMPING FROM 2 → 3 BEDROOMS

adds only ~$50

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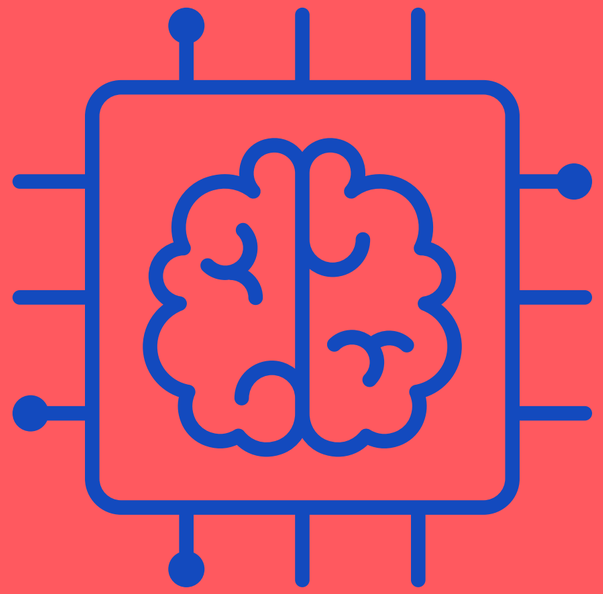
FROM INSIGHTS TO A REPEATABLE TOOL

Our Valuation Engine performs underwriting in less than 30 seconds.



FOR ANY PROPERTY, YOU ENTER...

- BEDROOMS
- GUEST CAPACITY
- DISTANCE TO THE MALL

THE TOOL RETURNS...

- PREDICTED NIGHTLY RATE (WITH ~85% ACCURACY)
- ESTIMATED MONTHLY REVENUE
- ESTIMATED ASSET SALE PRICE

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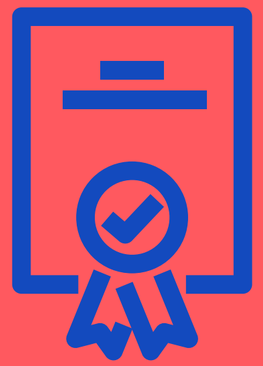
IMPACT SUMMARY

WITH THE CAPITAL INVESTOR, WE CAN:

REALLOCATE CAPITAL

INCREASE YIELD ON EXISTING ASSETS



STANDARDIZE UNDERWRITING

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THANK YOU.

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OUR TEAM